

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	pytest (unit tests) + manual response time measurement (Python `time` module)
Type of Testing	Functional Testing, Load Testing (manual), Unit Testing, Integration Testing
Target Module / API	Detection Engine (`src/detection/`), Gemini API Client (`src/gemini/client.py`),
Test Environment	Local (Windows 11, Python 3.10, 16GB RAM) + Streamlit Community Cloud (Free Tier)
Test Date	09 July 2026 - 12 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	Input	Expected Output	Result
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S.No	Test Scenario	Input	Expected Output	Result
1	Valid frustrated input	"I've been stuck on this	Emotion: Frustrated,	Pass
2	Valid curious input	"I wonder how neural	Emotion: Curious,	Pass
3	Short invalid input (< 10	"help"	Validation error message	Pass
4	Empty input	""	Validation error message	Pass
5	Very long input (2000	2000-char essay	Processed correctly, no	Pass
6	Mixed emotion input	"I'm a bit curious but also	Mixed emotion flag =	Pass
7	Keyword-boosted input	"I feel so bored and this	Rule applied = True,	Pass
8	Gemini API unavailable	Mock Gemini timeout	Fallback message	Pass
9	Concurrent users	5 browser tabs	All respond correctly	Pass
10	CSV logging	After any detection	New row appended to	Pass
			interaction_logs.csv	

Step 3: Unit Test Results

Test Suite: `tests/test\_preprocessing.py`

Test	Description	Result
`test_remove_html`	HTML tags removed from text	Pass
`test_remove_urls`	URLs stripped from text	Pass
`test_remove_mentions_hashtags`	@mentions and #hashtags removed	Pass
`test_normalize_whitespace`	Multiple spaces collapsed	Pass
`test_clean_text_full`	Full pipeline produces non-empty output	Pass

Test	Description	Result
`test_clean_for_bilstm`	BiLSTM path: lowercase, no stopwords	Pass
`test_clean_for_bert`	BERT path: preserves punctuation/case	Pass
`test_empty_input`	Empty string returns empty string	Pass

Test Suite: `tests/test\_detection\_pipeline.py`

Test	Description	Result
`test_rule_based_no_match`	No keywords rule not applied	Pass
`test_rule_based_match`	"stuck" frustrated boost applied	Pass
`test_ensemble_weights`	40/60 weights sum to 1.0	Pass
`test_ensemble_output_shape`	Output shape is (5,)	Pass
`test_mixed_emotion_detected`	Gap < 0.15 mixed = True	Pass
`test_single_emotion`	Gap > 0.15 mixed = False	Pass

Test Suite: `tests/test\_analytics.py`

Test	Description	Result
`test_log_creates_file`	log_interaction() creates CSV	Pass
`test_log_appends_row`	Second call appends, not overwrites	Pass
`test_aggregator_kpis`	KPIs computed correctly from mock data	Pass

Step 4: Performance Metrics

Metric	Value	Target	Status
Emotion Detection Response	~2.8 seconds	< 5 seconds	Pass
Gemini Guidance Generation	~1.5 seconds	< 5 seconds	Pass
End-to-End Response Time	~4.2 seconds	< 8 seconds	Pass
App Cold Start (Streamlit Cloud)	~12 seconds	< 30 seconds	Pass
CSV Log Write Time	< 0.1 seconds	< 1 second	Pass

Metric	Value	Target	Status
Analytics Dashboard Load Time	< 2 seconds	< 5 seconds	Pass

Step 5: Model Performance Metrics (Training Results)

Metric	BiLSTM	BERT	Ensemble
Training Accuracy	84.2%	89.7%	-
Validation Accuracy	81.3%	87.4%	88.1%
Test Accuracy	80.8%	86.9%	**87.6%**
Macro F1-Score	0.79	0.86	**0.87**
Inference Time (per sample)	45ms	120ms	165ms

Step 6: Issues Found & Resolved

S.No	Issue	Resolution
1	<code>`contractions`</code> library fails on Streamlit Cloud	Removed from requirements.txt; handled inline
2	Version conflicts in requirements.txt	Relaxed pinning to <code>`>=`</code> for cloud deployment
3	Gemini API key not found on cloud	Added interactive UI key input field
4	Remote push rejected (diverged history)	Resolved via <code>`git pull --rebase`</code>